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ECM-EFS: An ensemble feature selection based on enhanced co-association matrix

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ABSTRACT

Currently, feature selection faces a huge challenge that no single feature selection method can effectively deal with various data sets for all real cases. Ensemble learning is a potential promising solution to address this problem. We propose an ensemble feature selection method based on enhanced co-association matrix (ECM-EFS). Positive-co-association matrix (PCM), negative-co-association matrix (NCM), and relative-co-association matrix (RCM) are first introduced to discover the relationship among features by ensembling the results in multiple feature selection methods. To further produce a more stable feature selection result, *"Feature Kernel"* is also introduced and used as a starting point for feature selection. Comparative experiments with four state-of-the-art methods have confirmed that the ECM-EFS can provide more robust results. Moreover, compared with traditional ensemble feature selection methods, our method can compensate information loss and reduce computational cost significantly.

1. Introduction

As a fundamental topic of machine learning, feature selection has attracted extensive research interest, especially in the application of text processing and bioinformatics. Feature selection aims to remove redundant, irrelevant, and noisy features from the original feature space to select an effective feature subset. With respect to different selection strategies, feature selection methods fall into three categories, namely filter, embedded, and wrapper methods [1].

According to the *No Free Lunch Theorem* (NFL) [2], no single feature selection algorithm can achieve satisfactory performance on data sets that are generated in various scenarios. A single feature selection algorithm exhibits its own advantages but also has some limitations. Therefore, ensemble of multiple feature selection methods is considered a promising solution. Generally, an ensemble method combines the results of multiple methods into a single consensus result to reduce the risk of making a wrong choice. In addition, it is proven to be more stable than single feature selection methods.

Ensemble feature selection methods fall into three categories, namely, homogeneous, heterogeneous [3], and hybrid ensembles. A homogeneous ensemble divides the original data set into multiple data subsets and then uses the same feature selection method for feature selection in each data subset. Finally, it aggregates the generated feature subsets into a final feature subset [4]. A heterogeneous ensemble uses multiple feature selection methods to extract features from the same data set and combines the results by using a certain strategy [5]. The advantages of heterogeneous ensemble are twofold. First, it can improve the classification performance and comprehensively consider the results in multiple feature selection methods. Second, it can reduce the risk of making a wrong choice due to a single feature selection method. A hybrid ensemble integrates multiple data subsets, feature selection methods, and classifiers and uses the aggregate feature strategy to obtain the final results. Hybrid ensemble can achieve remarkable stability and highly accurate results. However, hybrid ensemble methods face highly complex combinatorial optimization problems.

The limitations of current ensemble feature selection methods can be summarized as follows.

(1) The relationship between features has not been deeply explored, which may result in ignoring the features that are strongly related to important features.
(2) The consensus strategy for ensemble feature selection mostly uses simple voting, which may miss some important features with minor votes.
(3) Most feature importance based evaluation methods use the voting number of the base selectors to make decision. These

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methods are prone to output biased results with significant deviations.

To address the above challenging issues, in this paper, we propose a new ensemble feature selection method based on enhanced co-association matrix. Although the traditional co-association matrix is reported to be used in clustering ensemble, no research work has been reported to apply it for ensemble feature selection as far as we know, because the traditional co-association matrix can not be directly applied to the selection of integration features. However, we have found the potential value of co-association matrix and redesigned it for the application for ensemble feature selection. For clarity, the main contributions of this paper are summarized as follows.

(1) We propose an ensemble feature selection based on enhanced co-association matrix. Three fine grained types are proposed, namely positive-co-association matrix (PCM), negative-co-association matrix (NCM), and relative-co-association matrix (RCM), which can not only enrich the information of the traditional co-association matrix, but also effectively reveal the relative relationship between features.
(2) A new consensus strategy based on enhanced co-association matrix is proposed. Instead of simple voting, our method fully considers all the results given by the base feature selectors, the importance of features and the relationship between features, and then gives the final feature selection results.
(3) A notion of *Feature Kernel*, which contains the most important features and also serves as the starting point for feature selection, is introduced. An algorithm is also presented to construct the feature kernel and select subsequent features starting from the feature kernel.
(4) An overall feature selection method namely ECM-EFS is proposed, and 14 data sets from UCI are used to verify the stability and robustness of the ECM-EFS. In addition, four state-of-the-art ensemble feature selection algorithms, namely, a homogeneous ensemble method [6], a heterogeneous ensemble method [7], a hybrid ensemble method [8] and Random Forest feature importance ranking [9], are compared with the proposal. Extensive experiments on a variety of data sets have demonstrated the effectiveness and efficiency of our method.

The remainder of this paper is organized as follows. Section 2 reviews the related work. Section 3 describes the proposed methodology in detail. Section 4 reports the empirical evaluation of the proposal and discusses the computational complexity of other ensemble selection methods. Finally, Section 5 concludes the paper.

2. Related work

Bagging [10] and boosting [11] are two well-known homogeneous ensemble methods that generate models on different distributions of the original data set. Saeys et al. [9] used bootstrap aggregation to generate 40 bags from the data. For each of the bags, a separate feature ranking was performed, and the ensemble was formed by aggregating the single rankings by weighted voting using linear aggregation. Davis et al. [12] determined the relevant feature list through feature subset selection method for each training set and then selected the features that appeared frequently in all feature lists to form the final feature list. Thomas et al. [13] selected different features according to various sub-sampling of the original data and then compared the stability of the features according to the Kuncheva index. The evaluation of these methods based on the number of times the features are selected or the order of the features in the feature lists obtained by the base feature selection methods. There may be multiple features with the same degree of importance, and only considering the order may leads to significant bias. Therefore, the method is one-sided and tends to lead to deviation.

Heterogeneous ensemble uses heterogeneous feature selection methods for the ensemble, thereby enabling a large number of optional feature selectors, such as instance level perturbation, feature level perturbation, stochasticity in the feature selector, Bayesian model averaging, or combinations of these techniques [14]. Aggregating the different feature selection results can also be achieved using different methods, such as weighted voting [15]. Brahim et al. [16] measured feature algorithm confidence and conflict with the other algorithms as an aggregation technique to assign a reliability factor that guided the final feature selection. In fact, the essence of these methods are to select features based on voting, which usually ignores the few but important results given in some base feature selection methods.

Hybrid ensemble generally has two forms, one is to integrate numerous data subsets with multiple feature selection methods, and the other is to integrate multiple feature selection methods with several classifiers. Tsymbal et al. [17] created ensembles of simple Bayesian classifiers with random subspaces and considered a hill climbing (HC)-based refinement cycle. However, the cost of computational complexity is high.

Currently, four commonly used integrated feature selection search strategies are available:(1) Hill Climbing (HC) [18]; (2) Genetic Ensemble Feature Selection (GEFS) [19]; (3) Ensemble Forward Sequential Selection (EFSS); (4) Ensemble Backward Sequential Selection (EBSS) [17].

Considering that the co-association matrix [20] constructed in the evidence accumulation model can be regarded as a similarity matrix, it can be used to depict the average frequency that a pair of points is partitioned into the same cluster. Therefore, it is often used as an important basis for dividing the basic partition in the clustering ensemble. Li et al. [21] used the co-association matrix to calculate the stability of the sample points and then partition the cluster structure. Zhong et al. [22] revealed more global information by refining the co-association matrix from the data points and basic cluster levels. Subsequently, they took a refined co-association matrix as an initial similarity matrix and transform it by path-based measure. Finally, it was applied to the visual assessment of cluster tendency [23]. Zhong et al. [24] also noticed that removing some negative evidence from the co-association matrix could improve the accuracy of clustering. A weighted co-association matrix [25] is proposed in clustering ensemble, where the weight is determined by some evaluation functions. Berikov et al. [26] provided a solution to the semi-supervised regression problem. The main idea is to compute the co-association matrix on labeled and unlabeled data as the similarity matrix in a regularization framework and use low-rank decomposition of the co-association matrix to accelerate the calculations and reduce the memory. Furthermore, Blakely et al. [27] used co-association matrix based on spectral clustering to complete phase identification task. Currently, the co-association matrix is widely investigated for clustering ensemble. However, works for feature selection are few.

3. Methodology

The ensemble feature selection method consists of two steps. The first step is to set the base feature selection method to perform feature selection on the data set and generate multiple feature lists. The second step is to design an consensus strategy to integrate multiple feature selection lists into the final feature selection list. Figure 1 shows the ECM-EFS framework. An enhanced co-association matrix is used to design an consensus strategy.

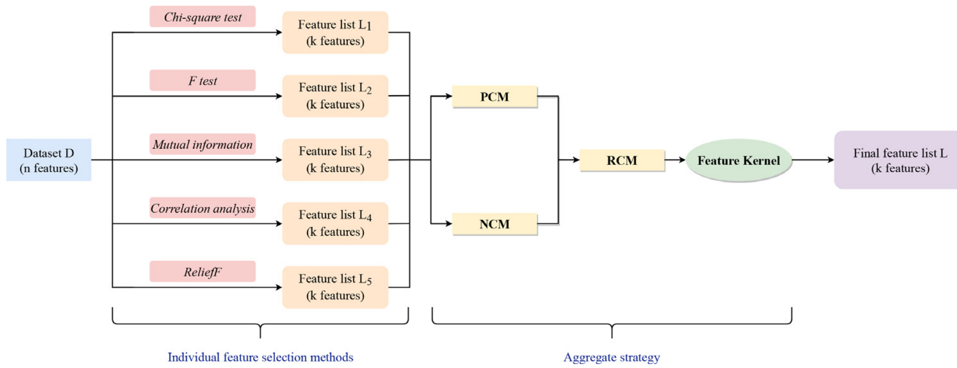
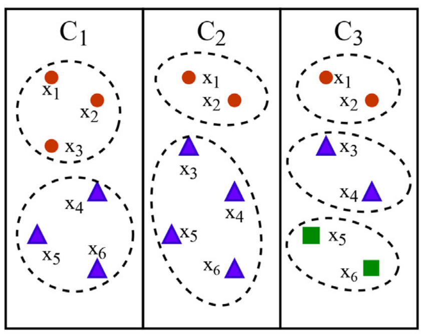


Fig. 1. The overview of ECM-EFS.



(a) Base clustering results

	x_1	x_2	x_3	x_4	x_5	x_6
x_1	1	1	1/3	0	0	0
x_2	1	1	1/3	0	0	0
x_3	1/3	1/3	1	2/3	1/3	1/3
x_4	0	0	2/3	1	2/3	2/3
x_5	0	0	1/3	2/3	1	1
x_6	0	0	1/3	2/3	1	1

(b) Co-association matrix

Fig. 2. An example of co-association matrix.

3.1. Co-association matrix

Let $X = \{x_1, x_2, \dots, x_N\}$ be a data set with N samples. Given a set of clustering algorithms $C = \{C_1, C_2, \dots, C_L\}$, according to clustering results $\prod = \{C_1, C_2, \dots, C_L\}$, the frequency of two sample points x_i and x_j in the same cluster can be obtained, all pairs of frequencies will form an $n \times n$ matrix, which is called the co-association matrix. Each element in the co-association matrix is calculated by:

$$PC_{ij} = \frac{1}{L}\left[\sum_{l=1}^{L} \mathbb{I}(C_l(x_i), C_l(x_j))\right] \tag{1}$$

$$\mathbb{I}(C_l(x_i), C_l(x_j)) = \begin{cases} 1, & C_l(x_i) = C_l(x_j), \\ 0, & C_l(x_i) \neq C_l(x_j), \end{cases} \tag{2}$$

where $C_l(x_i)$ indicates the label of x_i induced by clustering result C_l. Figure 2 gives a simple example of three clustering results $\prod = \{C_1, C_2, C_3\}$ on six samples.

The most significant difference between ensemble feature selection and clustering ensemble lies in the basic elements. The basic elements of clustering ensemble are the samples. Each element in the co-association matrix represents the frequency of two samples that appeared in the same cluster. The basic elements of ensemble feature selection methods are the features. Each sample must be clustered into different clusters for all clustering methods. While for an ensemble feature selection method, each feature is not necessarily to be selected in all feature selection methods. Traditional co-association matrix cannot accurately show the relationship among the features. Therefore, we propose PCM, NCM, and RCM. In addition, we propose *Feature Kernel* to construct a more effective initial feature subset.

Dataset	F_1	F_2	F_3	F_4
f_1	f_1	f_1	f_1	f_1
f_2	f_2	f_2	f_2	f_2
f_3	f_3	f_3	f_3	f_3
f_4	f_4	f_4	f_4	f_4
f_5	f_5	f_5	f_5	f_5
f_6	f_6	f_6	f_6	f_6

Fig. 3. An example of base feature selection.

3.2. PCM and NCM

Let $X = \{f_1, f_2, \dots, f_m\}$ be a data set that contains m features and $\prod = \{F_1, F_2, \dots, F_L\}$ be the result of L feature selection methods. The result of feature selection by using different feature selection methods is the initial matrix with the size of $m \times (L+1)$, where L represents the number of feature selection methods. Each column represents a feature selection result, and the ith row indicates the selecting results in the ith feature. The last column corresponds to the voting column, which indicates the number of times each feature is selected. Obviously, the more votes a feature receives, the more times it is selected.

Figure 3 provides a simple example of the results in the four feature selection methods $\prod = \{F_1, F_2, F_3, F_4\}$ on six features. Taking Fig. 3 as an example, the initial matrix in Fig. 4 can be obtained.

If a feature is not selected by a method, then we regard the method as abstaining for this feature. The relationship between a pair of features f_i and f_j regarding the method F_l can be summarized into four cases: combination, repulsion, independence and abstention. The first type is if f_i is selected, then f_j must be selected. The second type is if f_i is selected, then f_j should not be

	F_1	F_2	F_3	F_4	votes
f_1	1	1	0	1	3
f_2	1	0	1	0	2
f_3	0	1	1	0	2
f_4	0	0	1	1	2
f_5	1	1	1	1	4
f_6	1	1	0	1	3

Fig. 4. Initial matrix.

selected. The third type means that f_i and f_j are not related and thus are independent of each other. The last type means that neither f_i nor f_j has ever been chosen, which in turn inferred that the relationship between these two features is worthless and meaningless. Therefore, we do not consider the last type.

$$(F_l(f_i), F_l(f_j)) = \begin{cases} (1,1), & F_l \text{ selects both } f_i \text{ and } f_j, \\ (1,0), & F_l \text{ selects} f_i \text{ but not } f_j, \\ (0,1), & F_l \text{ selects} f_j \text{ but not } f_i, \\ (0,0), & F_l \text{ doesn't select } f_i, f_j \\ & \text{and it is regarded as abstention} \end{cases} \tag{3}$$

To describe the degree to which the features tend to be combined, the PCM is proposed. Each element in PCM corresponds to the degree to which two features should be combined, and it is defined as follows:

$$PF_{ij} = \frac{v}{L} \times \left[\frac{1}{v} \sum_{l=1}^{L} \mathbb{I}(F_l(f_i), F_l(f_j)) \right] \tag{4}$$

where v is the number valid votes. The number of valid votes is the number of non-abstaining votes. The first term in Eq. (4) represents the ratio of valid votes to the total number of votes, and the second term represents a measure of the relationship between a pair of features in the number of valid votes. $\mathbb{I}(F_l(f_i), F_l(f_j))$ is defined as follows:

$$\mathbb{I}(F_l(f_i), F_l(f_j)) = \begin{cases} 1, & F_l(f_i) = F_l(f_j) = 1, \\ 0, & \text{others}, \end{cases} \tag{5}$$

Taking the initial matrix in Fig. 4 as an example, the PCM can be obtained (Fig. 5(a)).

Apparently, PCM is a symmetric matrix with the diagonal elements of 1. However, only considering the closeness between features cannot accurately reflect the relationship between features. Thus, we further construct the NCM, which reflects the degree of repulsion between a pair of features. Each element in NCM indicates the degree to which the two features should be separated, that is, the degree of f_j should not be selected together with f_i.

Each element in the NCM is also defined as Eq. (4). However, the definition of $\mathbb{I}(F_l(f_i), F_l(f_j)$ is different from that of PCM. The example in Fig. 4 shows that NCM can be obtained as shown in Fig. 5(b).

$$\mathbb{I}(F_l(f_i), F_l(f_j)) = \begin{cases} 1, & F_l(f_i) \neq F_l(f_j), \\ 0, & \text{others}, \end{cases} \tag{6}$$

3.3. RCM

Based on PCM and NCM, we propose the concept of RCM to integrate these two types of co-association matrices. RCM reflects the strength of the relative relationship between features from two aspects: the relative relationship between each two features, namely, combination, repulsion, and independence, and the relative relationship between each feature and other features.

First, the RCM_1 is constructed. Each element in the RCM_1 represents the relative relationship between features, which is called the relative-co-association value. A positive value indicates that the two features tend to be combined. Meanwhile, a negative value indicates that the features tend to be repulsive. If the value is zero, then the features tend to be independent. The greater the absolute value of the element value is, the stronger the relationship between the features.

$$RCM_1 = PCM - NCM \tag{7}$$

To clearly indicate the degree of clarity of the relationship between each feature and the rest of the features, a column is added to RCM_1, namely, the co-association value ($coval_{fi}$), and the calculation method is given by Eq. (8). The larger the value of $coval_{fi}$, the more obvious the relationship between this feature and other features will be. Therefore, we obtain a $(L \times (L+1))$ matrix RCM_2. Then, we add a voting column to RCM_2 to indicate the importance of each feature in all the feature selection methods used for ensembling. Thus, the final $RCM(L \times (L+2))$ is obtained. Figure 6 shows the RCM of the example in Fig. 4.

$$coval_{fi} = \sum_{j=1}^{L} |f_i, f_j| \tag{8}$$

3.4. Feature kernel

The initial choice of features is crucial for the effectiveness of the feature selection result. Therefore, we introduce "*Feature Kernel*" which contains the most important features that should be selected. The importance is measured by two aspects: the number of votes and the co-association value.

In RCM, the last column contains the votes of each feature, which can be considered the importance of the feature. The more votes a feature obtains, the more important it is considered in the feature selection process. The co-association value column in the RCM matrix indicates the strength of the relationship between each feature and the remaining features. Therefore, we sort the features in descending order according to the number of votes. The features for the cases with the same number of votes are sorted in descending order according to the co-association value. Finally, we select the first $\lfloor k/3 \rfloor$ features to form a "*Feature Kernel*", where k is the final number of features to be selected. Algorithm 1 provides the pseudocode for generating feature kernel instances.

3.5. ECM-EFS

We proposed an ensemble feature selection method based on enhanced co-association matrix(ECM-EFS), that is, PCM, NCM, and RCM. The overall framework of ECM-EFS is shown in Fig. 1. PCM can reflect the degree to which the relationship between features tends to be combination. NCM can reflect the degree to which the relationship between features tends to be repulsion. RCM can comprehensively reflect that the relationship between features tends to be combination, repulsion, or independence. Feature kernel contains the most important features, and these features are the starting point for feature selection. Finally, based on the results of PCM, NCM, RCM and feature kernel, we give the complete steps for feature selection by ECM-EFS.

	f_1	f_2	f_3	f_4	f_5	f_6
f_1	1	1/4	1/4	1/4	3/4	3/4
f_2	1/4	1	1/4	1/4	1/2	1/4
f_3	1/4	1/4	1	1/4	1/2	1/4
f_4	1/4	1/4	1/4	1	1/2	1/4
f_5	3/4	1/2	1/2	1/2	1	3/4
f_6	3/4	1/4	1/4	1/4	3/4	1

(a) Positive-co-association matrix

	f_1	f_2	f_3	f_4	f_5	f_6
f_1	0	3/4	3/4	3/4	1/4	0
f_2	3/4	0	1/2	1/2	1/2	3/4
f_3	3/4	1/2	0	1/2	1/2	1/2
f_4	3/4	1/2	1/2	0	1/2	3/4
f_5	1/4	1/2	1/2	1/2	0	1/4
f_6	0	1/4	1/2	3/4	1/4	0

(b) Negative-co-association matrix

Fig. 5. Information matrix.

	f_1	f_2	f_3	f_4	f_5	f_6	$coval_{fi}$	votes
f_1	1	-1/2	-1/2	-1/2	1/2	3/4	3.75	3
f_2	-1/2	1	-1/4	-1/4	0	-1/2	2.5	2
f_3	-1/2	-1/4	1	-1/4	0	-1/4	2.25	2
f_4	-1/2	-1/4	-1/4	1	0	-1/2	2.5	2
f_5	1/2	0	0	0	1	1/2	1	4
f_6	3/4	-1/2	-1/4	-1/2	1/2	1	3.5	3

Fig. 6. Relative-co-association matrix (RCM).

Algorithm 1 Feature kernel.

Require: PCM;NCM;
Ensure: feature kernel
calculating the RCM_1;
calculating the RCM_2;
calculating the RCM;
All features are arranged in descending order according to the number of votes obtained, and the feature sequence FS_1 is obtained;
Grouping according to the number of votes, the features with the same number of votes are sorted in descending order according to the co-association value, and the feature sequence FS_2 is obtained;
Taking the first $\lfloor k/3 \rfloor$ features in FS_2 to form a feature kernel;

Suppose there are m features in the original feature set $F = \{f_1, \ldots, f_m\}$ in X, and k features are to be selected to form the final feature subset $F' = \{f'_1, \ldots, f'_k\}$. The overall process of ECM-EFS to perform feature selection is as follows:

Step 1: The features in *Feature Kernel* are added to the F' in the order in which they are added to the *Feature Kernel*. That is, $F' = \{f'_1, \ldots, f'_{\lfloor k/3 \rfloor}\}$.

Step 2: Select the feature with the largest relative-co-association value in turn according to the features in F', until F' contains k features. Assuming that the selection is based on f'_i in F', there will be three situations in this process:

1. The feature with the largest relative-co-association value with f'_i is also in F', then ignore this feature, reselect and add the selected feature (not appearing in F') to F'.
2. If there are more than one features with the equal and largest relative-co-association value with the feature f'_i, then the one with the highest number of votes is selected first. If the vote number of more than one features are same, then the one with the largest co-association value with the feature f'_i will be selected and added to F'.
3. If there is only one feature with the largest relative-co-correlation value with f'_i, and this feature is not in F', then add this feature to F'.

We also perform a theoretical analysis of ECM-EFS in terms of its computational cost. The corresponding time complexity ECM-EFS is estimated as follows:

$$T_{\text{ECM-EFS}} = T_{\text{matrices}} + T_{FK} + T_{FSP} \tag{9}$$

where T_{matrices}, T_{FK}, and T_{FSP} denote the computational costs for the calculation of multiple matrices, construction of *feature kernel*, and feature selection process, respectively. Specifically, T_{matrices} is estimated as follows:

$$T_{\text{matrices}} = T_{InitMtx} + T_{PCM+NCM} + T_{RCM} \tag{10}$$

Given a data set that contains m features, if k features are selected, then the time complexity of constructing the initial matrix is $O(m^2)$. According to the relationship between each pair of features in the initial matrix, *PCM* and *NCM* can be constructed, so the complexity of calculating *PCM* and *NCM* is $O(m^2)$. RCM_1 is obtained by subtracting *PCM* and *NCM*, and the computational complexity of getting RCM_1 is $O(m^2)$ as the size of PCM and RCM are both m^2. According to Eq. (8), the complexity of calculating the co-association value is $O(m^2)$. Therefore, the computational complexity of constructing *RCM* is $O(m^2)$.

The *feature kernel* is mainly determined by the voting column in *RCM*. If the number of votes is fixed, then the co-association value would be compared, so the complexity of constructing the *feature kernel* is only $O(m)$, that is, $T_{FK} = O(m)$. In the feature selection process, if the first case occurs in the search process, each of the remaining features can be determined by looking up the relative-co-association value in *RCM* in turn. Because the *feature kernel* already contained $\lfloor k/3 \rfloor$ features, the complexity of feature selection starting from the *feature kernel* is $O(m)$. Therefore, the total computational complexity of ECM-EFS is $O(m^2)$.

4. Experiment

The experimental setup is introduced in Section 4.1. We apply the *Wilcoxon Signed Rank Test* to prove the robustness of our method. The experimental results are presented in Section 4.2. Finally, we compare our proposal with four state-of-the-art ensemble methods in Section 4.3 and analyze their computational complexity in Section 4.4.

4.1. Experimental setting

A total of 14 data sets from the UCI machine learning repository are selected in our experiment. We not only selected binary

Table 1
Data sets used in the experiments.

Data sets	Instances	Features	Classes	Number
Breast Cancer Wisconsin	569	31	2	D1
glass	214	9	7	D2
Ionosphere	351	34	2	D3
mushroom	8124	23	2	D4
ParkinsonsTelemonitoring	5857	21	42	D5
QSAR biodregation	1055	41	2	D6
Segmentation	2130	19	7	D7
Wine	173	13	3	D8
Zoo	101	16	7	D9
Soybean	307	35	19	D10
HillValley	1212	109	2	D11
LIBRASMovement	360	91	15	D12
semeion	1593	256	10	D13
qsar androgen receptor	1687	1024	2	D14

Table 2
Individual feature selection algorithms information.

Method	Description
Chi-square test (χ^2 test) [28]	It calculates the correlation between the features and the label.
F test [29]	It estimates the degree of linear correlation between the feature variables and the labels.
Mutual information (MI) [30]	The features are ranked on the basis of mutual information between the features and the labels, and the least relevant feature variables are filtered out
Correlation analysis (CA) [31]	Pearson's correlation coefficient is used to test the linear correlation between features and labels.
ReliefF [32]	It measures degree of distinction between samples of the same class and samples of different classes.

classification data sets but also multi-class data sets. In addition, the data sets contain balanced and imbalanced data sets. The main characteristics of the data sets, including the numbers of instances, features, and classes, are given in Table 1.

We select five feature selection algorithms for the ensemble of ECM-EFS. Table 2 shows the information of them.

To evaluate the classification performance of the data subset selected by the feature selection method, we applied the support vector machine (SVM) as classifier. SVM performs classification by looking for the optimal classification hyperplane in a fixed data space. Feature selection directly affects the structure of the data space, thus, it is sensitive to the feature selection. For each class, a SVM is designed to classify it from all the other classes. Therefore, k SVMs will be constructed for k classes. At the testing phase, a testing sample will be classified into the class with the largest classification function value. We selected a linear kernel for SVM and set $\sigma = 0.1$.

To evaluate the performance of ECM-EFS in classification tasks, we used accuracy (Acc), precision (P), recall (R), F1 score (F1), and *G*-mean as the performance metrics, which are widely used for feature selection studies. In order to ensure the validity of the experimental results, we carried out the experiment by 10-fold cross-validation.

The following metric expressed as follows.

$$Acc = \frac{TP+TN}{TP+TN+FP+FN}, P = \frac{TP}{TP+FP}, R = \frac{TP}{TP+FN} \tag{11}$$

For imbalanced data sets, *G*-mean is introduced as an evaluation criteria [33].

$$F1 = \frac{2\times P\times R}{P+R}, G\text{-mean} = \sqrt{\frac{TP}{TP+FN}\times\frac{TN}{TN+FP}} \tag{12}$$

where, TP is the number of positive instances correctly classified as positive. TN is the number of negative instances correctly classified as negative. FP is the number of the negative instances incorrectly classified as positive. FN is the number of the positive instances incorrectly classified as negative.

4.2. Comparing with single feature selection methods

For a data set that contains m features, we select $k = \lfloor m/2 \rfloor$ features based on experience. Figure 7 shows the performance of the compared methods. The results in ECM-EFS can be summarized with two folders. (1) ECM-EFS achieved the best performance. For example, it obtained the highest accuracy for five data sets: ParkinsonTelemonitoring, Wine, Zoo, Soybean, and HillValley; (2) The performance measures of ECM-EFS is neither the best nor the worst. Taking the Acc metric as an example, the accuracies of ECM-EFS for the nine remaining data sets are not the highest but also not the lowest. The main reason for the first case is that the distributions of these data sets are relatively complex. A single feature selection algorithm usually cannot cope with such complex cases well. However, ECM-EFS can consider the selection results of all methods and provide more comprehensive options, thereby resulting in high performance. As a set of parameters for ECM-EFS are usually suitable for a certain type of data sets, ECM-EFS cannot achieve the best performance for all the data sets.

To further test the stability of ECM-EFS, we launch rank testing for the comparison of the experimental results. The indicators obtained by all feature selection methods for each data set are sorted in ascending order. The position obtained by each method is called "*Rank*". The larger the *Rank* of a method is, the better performance the method will exhibit [34]. Specifically, as shown in Fig. 8(a), w.r.t. the Accuracy scores. Combined with 14 comparisons, the rank sum value obtained by ECM-EFS is the largest, implying that the performance of ECM-EFS is generally better. As shown in Fig. 8(b), ECM-EFS are ranked in the top 2 in 10 comparisons, thereby showing that ECM-EFS has high robustness and stability.

To test the robustness of the proposal, we apply *Wilcoxon signed rank test* [35] on the results of the selected data sets. It is used for pairwise comparison between the two algorithms. First, the null hypothesis of no significant difference between the two methods is defined. The differences between the two compared methods are calculated, and the experimental results are ranked. Then, based on the ranked results, the positive differences are sumed as R^+, and the negative differences are summed as R^-. If the difference between R^+ and R^- is sufficiently large, then a significant difference between the two methods is observed, and the null hypothesis is rejected. In addition, according to p-values that are compared with the given significance level α, if p-value $\le \alpha$,then the null hypothesis is rejected. Table 3 shows the *Wilcoxon signed rank test* results of performance measures.

For comparisons with CA and ReliefF, the p-values are far lower than the standard significance level 0.05, which leads to the conclusion that ECM-EFS is significantly better than the two other methods. For the comparison with MI and F test, not only the p-values are lower than the standard confidence level of 0.1, but the differences between R^+ and R^- are also larger, which suggest the significant differences in the comparison results. For the case with χ^2 test, the p-value in accuracy are far lower than the standard significance level 0.05, and the R^+ is apparently higher than R^- for all the performance metric. Therefore, combining all the results together, the null hypothesis should be rejected, and ECM-EFS is significantly better than the compared algorithms on the all selected data sets.

To verify the effectiveness of the feature kernel, we first performed a set of experiments without setting the feature kernel(NFK), and then with the feature kernel(FK). The experimental results are shown in Table 4.

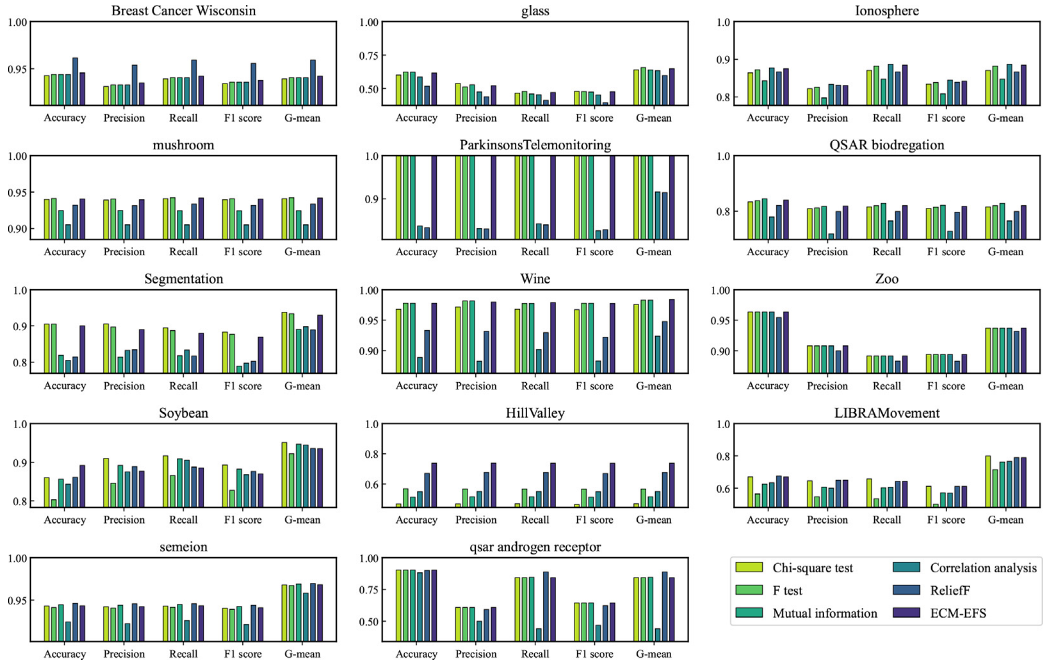
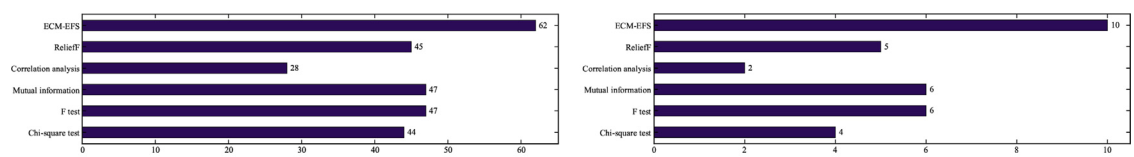


Fig. 7. Performance of each dataset in different feature selection methods.



(a) Sum of ranking times (b) Number of times being ranked in top 2

Fig. 8. The number of times that each method is ranked (a) the sum and (b) the top two among all comparisons in the Accuracy metric.

Table 3
Wilcoxon signed rank test of ECM-EFS versus the compared methods.

Accuracy				Precision				Recall			
ECM-EFS vs.	R^+	R^-	*p*-value	ECM-EFS vs.	R^+	R^-	*p*-value	ECM-EFS vs.	R^+	R^-	*p*-value
χ^2 test	59	7	1.12E−02	χ^2 test	39	27	3.12E−01	χ^2 test	39	27	3.12E−01
F test	50	16	7.12E−02	*F* test	65	13	2.27E−02	*F* test	59	19	6.30E−02
MI	63	15	3.26E−02	MI	66	25	8.11E−02	MI	68	23	6.21E−02
CA	89	2	1.32E−03	CA	88	3	1.70E−03	CA	84	7	4.00E−03
ReliefF	93	12	4.27E−03	ReliefF	86	19	7.16E−02	ReliefF	85	20	2.09E−02

F1 score				*G*-mean			
ECM-EFS vs.	R^+	R^-	*p*-value	ECM-EFS vs.	R^+	R^-	*p*-value
χ^2 test	40	26	2.82E−01	χ^2 test	42	24	2.25E−01
F test	62	16	3.88E−02	*F* test	58	20	7.34E−02
MI	69	22	5.40E−02	MI	70	21	4.67E−02
CA	88	3	1.70E−03	CA	86	5	2.60E−03
ReliefF	91	14	6.70E−03	ReliefF	82	23	3.38E−02

The bold numbers in Table 4 indicate that this method achieved better performance. On average, the group with feature kernel wins for 9 times and loses for 5 times in the comparisons. In particular, adding feature kernel can significantly improve the accuracy. The results suggest that feature kernel is effective for most cases. Especially for higher-dimensional datasets, such as LIBRASMovement, semeion and qsar androgen receptor, the performance has been significantly improved after adding feature kernel. It can be seen that the more features the data set contains, the more important a set of core features is needed for feature selection, and feature kernel can identify the core features. The losing cases imply that feature kernel still has weakness. If some important features were not selected by most single feature selection algorithms, then these features cannot receive a high number of votes or obtain a low co-association value. These features may be selected before the feature kernel was set. However, feature kernel sets, a more stringent condition, may lead to the discarding of such features. Therefore, the accuracy of Segmentation and HillValley are reduced after the feature kernel is set. The reason why the accuracy of Ionosphere and Wine is equal under the conditions of NFK and FK is that their important features are easily identified, and the base feature selection methods give the same selection result, so no matter whether the feature kernel is added or not, the final selected features are the same. As a whole, feature kernel can effectively improve the robustness of feature selection.

Table 4
The experimental results of set feature kernel.

Data sets	Accuracy		Precision		Recall		F1 score		*G*-mean	
	NFK	FK	NFK	FK	NFK	FK	NFK	FK	NFK	FK
Breast Cancer Wisconsin	0.9439	**0.9456**	0.9328	**0.9349**	0.9405	**0.9421**	0.9358	**0.9376**	0.9405	**0.9421**
glass	0.6091	**0.6159**	**0.5292**	0.5213	0.4646	**0.4710**	0.4752	**0.4768**	0.6411	**0.6488**
Ionosphere	**0.8750**	**0.8750**	0.8255	**0.8301**	**0.8889**	0.8846	**0.8436**	0.8419	**0.8889**	0.8846
mushroom	0.9135	**0.9405**	0.9143	**0.9396**	0.9137	**0.9417**	0.9134	**0.9402**	0.9137	**0.9417**
ParkinsonsTelemonitoring	0.9990	**1.0000**	0.9990	**1.0000**	0.9991	**1.0000**	0.9990	**1.0000**	0.9995	**1.0000**
QSAR biodregation	0.8378	**0.8396**	0.8121	**0.8183**	0.8199	**0.8199**	0.8137	**0.8173**	0.8199	**0.8199**
Segmentation	**0.9048**	0.9000	**0.8969**	0.8898	**0.8870**	0.8799	**0.8769**	0.8693	**0.9336**	0.9293
Wine	**0.9778**	**0.9778**	**0.9815**	0.9796	0.9775	**0.9789**	**0.9777**	0.9774	0.9828	**0.9839**
Zoo	0.9000	**0.9636**	0.7505	**0.9083**	0.7209	**0.8917**	0.7271	**0.8943**	0.8352	**0.9371**
Soybean	0.8548	**0.8919**	**0.9005**	0.8769	**0.9146**	0.8855	**0.8877**	0.8700	**0.9496**	0.9357
HillValley	**0.8852**	0.7377	**0.8874**	0.7384	**0.8840**	0.7376	**0.8848**	0.7370	**0.8840**	0.7376
LIBRASMovement	0.6361	**0.6694**	0.6175	**0.6497**	0.6111	**0.6425**	0.5822	**0.6107**	0.7680	**0.7889**
semeion	0.8706	**0.9433**	0.8704	**0.9423**	0.8758	**0.9434**	0.8672	**0.9411**	0.9291	**0.9682**
qsar androgen receptor	0.8580	**0.90236**	**0.6691**	0.6083	**0.8576**	0.84256	**0.7086**	0.64365	**0.8576**	0.84256

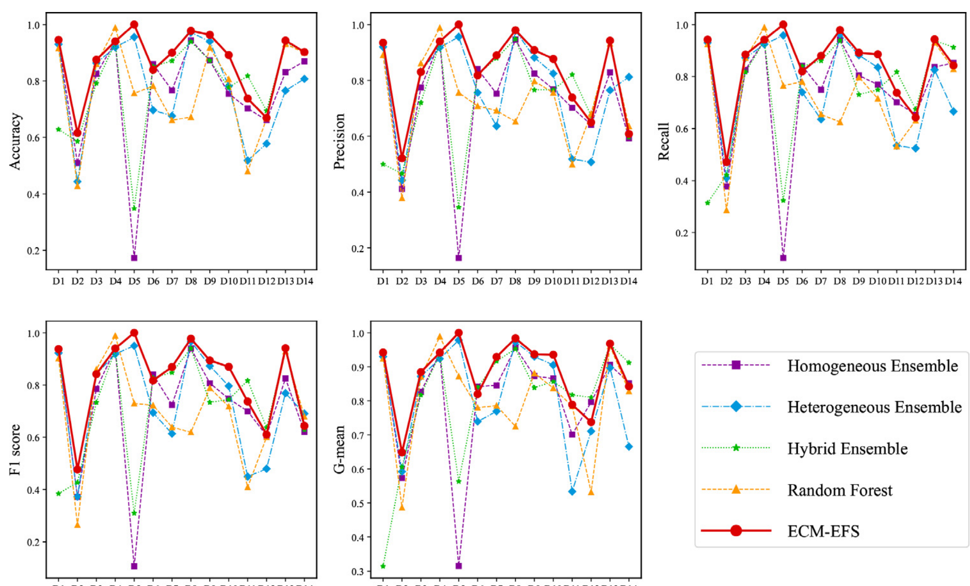


Fig. 9. Performance on different ensemble feature selection methods.

4.3. Compared with ensemble feature selection methods

To further validate the performance of our proposal, we compare ECM-EFS with four state-of-art ensemble feature selection methods, including a homogeneous ensemble [6], a heterogeneous ensemble [7], a hybrid ensemble feature selection method [8], and Random Forest feature importance ranking [9]. For the fairness of the comparisons, we conducted the experiments using the same selected data sets, the same single feature selection method for the first three methods, and the same classifier.

Figure 9 represents the performance of the five compared ensembled feature selection methods. Experimental results show that the performance of ECM- EFS on all selected data sets are better than those of the ensemble feature selection method proposed by Pes et al. [6] and Bolón-Canedo et al. [7]. ECM-EFS has also been observed to defeat the method proposed by Zhang et al. [8] for most of the selected data sets. The method proposed by Zhang et al. [8] performed better in high-dimensional data, but the method had a very high computational cost and time overhead because of the hybrid ensemble approach. Apparently, ECM-EFS significantly outperforms Random Forest feature importance ranking. From a global view point, the performance of ECM-EFS in terms of accuracy, precision, recall, F1 score, and *G*-mean is better than the compared ensemble methods.

4.4. Discussion of computational complexity

To further compare the performance of our proposal and the first three selected ensemble feature selection methods, we set out to analyze their computational complexities in detail. Without loss of generality, we assume that the average complexity of all feature selection methods is $O(p)$, and the complexity of classification is $O(q)$.

For a data set with N samples, the computational complexity of the method proposed by Pes et al. [6] is composed of four parts. (1) In terms of sampling, the computational complexity is $O(MN) + O(MKN)$; (2) In terms of feature selection, the computational complexity of is $O(MKp)$; (3) In terms of aggregation strategy, the first aggregation will generate M feature lists. If each feature list contains k features, then the computational complexity of the first aggregation is $O(M(K+1)k)$. The second aggregation integrates the M feature lists into a final feature list, and its complexity is $O((M+1)k)$; (4) The computational complexity in classification is $O(q)$.

The method proposed by Bolón-Canedo et al. [7] uses five filter methods. Thus, the computational complexity in the feature selection is $O(5p)$. As the classification procedure is executed on each of the five data subsets, the computational complexity of the classification is $O(5q)$. 10-fold cross-validation is applied for

Table 5
Complexity analysis of ensemble feature selection methods.

Method	Sample	Feature selection	Classifier	Aggregation strategy
Pes et al. [6]	$O(MKN)$	$O(MKp)$	$O(q)$	$O(M(K+1)k+(M+1)k)$
Bolón-Canedo et al. [7]		$O(5p)$	$O(5q)$	$O(50N)$
Zhang et al. [8]	$O(MN)$	$O(5Mp)$	$O(q)$	$O(Mk)$ Hash search $O(Mk\log L)$ Binary search
ECM-EFS		$O(5p)$	$O(q)$	$O(m^2)$

the prediction, and the simple voting is used to obtain the final results. Therefore, the time complexity of aggregating sample label results is $O(50N)$.

The computational complexity of the method proposed by Zhang et al. [8] consists of two parts. In the homogeneous ensemble stage, the sampling complexity of this method is $O(MN)$. The time complexity of the five feature selection methods for feature selection is $O(5Mp)$. For the heterogeneous ensemble stage, they first evaluate the similarity among the feature subsets. The evaluation process involves looking up for all the features that appear in feature lists. For a list with the length of L, if the hash search is used in the search process, then the time complexity is $O(1)$. If the binary search is used in the search process, then the time complexity is $O(\log L)$. Therefore, the time complexity of this stage is $O(Mk)$ or $O(Mk\log L)$.

The complexity of the comparison results of the ensemble feature methods are shown in Table 5. The time complexity of ECM-EFS is the lowest in terms of sampling, feature selection, classification, and aggregation strategies, thereby suggesting that our proposal can achieve the best time performance.

5. Conclusion

In this paper,we propose ECM-EFS method to address the problem of unstability and poor robustness of feature selection methods. The relative relationship between features is explored through PCM, NCM, and RCM, and the most important features are explored by constructing feature kernel, which are used as the starting point for feature selection. Therefore, ECM-EFS is stable and high feature selection performance. The experimental comparisons between ECM-EFS and other single feature selection methods on 14 data sets verified the performance. Compared with other ensemble feature selection methods, ECM-EFS can also get high performance with lower computational complexity.

However, for some data with special structures, such as manifold data, our proposal cannot work well. Furthermore, ECM-EFS cannot automatically obtain the optimal number of features. These issues will lead to our studies in the future.

Declaration of Competing Interest

Authors declare that they have no conflict of interest.

Data availability

The authors do not have permission to share data.

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Supplementary material

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